

Modeling Artist Lifecycle Sustainability and Decline in Global Music Streaming

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Abstract—This paper describes our attempt to predict artist popularity decline on Spotify using career trajectory features. We built an artist-year aggregation pipeline from 32,000 track records (1,094 artists, 1990–2019) and trained three classifiers under a time-aware evaluation framework. Our best model, Random Forest, achieves PR-AUC = 0.201 against a random baseline of 0.095—a $2.1\times$ improvement that, while statistically real, is below what we initially hoped for. We also used K-Means clustering to identify four artist lifecycle archetypes, validated by chi-square testing ($p < 0.0001$). We are honest that both results fall short of what would be needed for real-world use, and we identify concrete improvements for future work.

Index Terms—Artist lifecycle, decline prediction, Spotify, PR-AUC, imbalanced classification, K-Means clustering.

I. INTRODUCTION

Spotify popularity scores reflect recent streaming activity and can decay quickly for artists who stop releasing music or lose playlist placement. Predicting which artists are heading toward a decline could help music labels decide where to direct their support and resources.

This project has three parts: (1) building a data pipeline that turns track-level Spotify data into artist-year career profiles; (2) training classifiers to predict whether an artist will decline in the next year; and (3) using clustering to find common career patterns. We were hoping for stronger predictive performance. This report is honest about what we achieved and what we would do differently.

II. RELATED WORK

Most music information retrieval research predicts track-level properties—genre, mood, hit potential—from audio descriptors [3]. Celma [3] showed that streaming platforms strongly favor already-popular artists, creating structural momentum that our trajectory features aim to capture. Askin and Mauskopf [4] found that chart success depends on how similar a song is to existing hits, suggesting audio fit matters for initial popularity but not necessarily for career longevity. Zackery studied the "sophomore slump"—the tendency to underperform after a strong debut—which is the closest prior work to our setting. Unlike these studies, we model multi-year artist trajectories rather than individual song outcomes, and we use career momentum features rather than audio descriptors alone.

III. DATA AND METHODOLOGY

A. Dataset

We used *spotify_songs_JA.csv* [5], containing ~32,000 tracks with artist names, release dates, popularity scores (0–100), and 12 audio features. After filtering for valid records and release years ≥ 1990 , we kept 26,304 tracks. We then removed artists with fewer than three active years, leaving

3,474 artist-year rows across 1,094 artists.

B. Features

We engineered 9 trajectory features from the popularity history:

TABLE I: Engineered Trajectory Features

Feature	What it measures
peak_popularity	Highest popularity the artist ever reached
gap_from_peak	How far below their peak they currently are
years_since_peak	How long ago they were at their best
career_slope	Overall upward or downward trend over career
release_consistency	Whether they released music the prior year
yearly_change	Year-over-year popularity change
variability	How much popularity fluctuates year to year
median_popularity	Typical popularity, ignoring outlier tracks
max_popularity	Best single track in that year

We also included 6 audio feature means (danceability, energy, valence, tempo, acousticness, speechiness), giving 19 features total.

C. Decline Label

We labeled a year as "decline" if the artist's mean popularity dropped ≥ 15 points the following year. We chose 15 because it is above the typical year-to-year noise (median swing = 12 pts) but below one standard deviation ($\sigma = 23$ pts). This produced a class prevalence of 8.7%—the data is imbalanced, with only 1 in 11 artist-years being a true decline.

D. Classifiers

We evaluated three classifiers, each chosen for a specific reason. The **Decision Tree** provides interpretable if-then rules a domain expert can audit. The **Random Forest** reduces variance by averaging 200 decorrelated trees, producing calibrated probabilities suitable for threshold tuning. **Gradient Boosting** (HistGBM) iteratively corrects residuals and natively handles missing values—important for first-year artists where expanding features are undefined. All three use `class_weight='balanced'` so the 8.7%-minority decline class is

not ignored in favor of the majority.

E. Evaluation

We split data by time: training on years ≤ 2016 , testing on 2017–2019. We also used **expanding-window cross-validation** (4 folds, test years 2015–2018) for hyperparameter tuning—each fold trains on all data before the test year, mimicking real deployment. Standard k-fold is invalid here because it would let future data leak into training. Because the data is imbalanced, we used **Precision-Recall AUC (PR-AUC)** as the primary metric. A model that randomly guesses achieves PR-AUC = 0.095 (equal to the class prevalence), so any score above that represents genuine learned signal.

IV. RESULTS

A. Hyperparameter Tuning

Grid search over the expanding-window CV folds, scored on PR-AUC, selected the following best parameters:

TABLE II: Hyperparameter Tuning via Expanding-Window CV

Model	Best CV PR-AUC	Key Parameters
Random Forest	0.321	n_estimators=200, max_depth=5, min_samples_leaf=10
Gradient Boosting	0.302	max_iter=100, max_depth=3, lr=0.05, min_samples_leaf=10

B. Holdout Test Results

TABLE III: Holdout Test Results (2017–2019, threshold = 0.5)

Model	PR-AUC	ROC-AUC	Recall	Accuracy
Decision Tree	0.163	0.678	0.734	0.537
Random Forest	0.201	0.702	0.772	0.526
Gradient Boosting	0.182	0.681	0.772	0.509
Random baseline	0.095	0.500	—	—

All three models beat the random baseline. Random Forest is best (PR-AUC = 0.201, 2.1 $\times$ baseline). The high recall (0.77) means the model catches most real declines, but the low precision (0.14 at default threshold) means it also raises many false alarms.

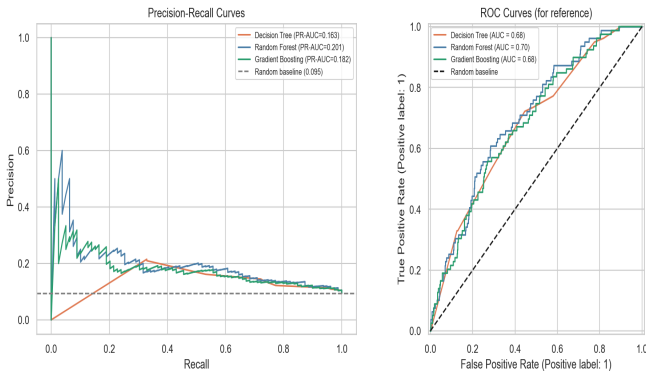


Fig. 1. PR and ROC curves on the holdout test set. Dashed line = random PR-AUC baseline (0.095).

C. Threshold Tuning

The default threshold of 0.5 is arbitrary. We derived an F1-optimal threshold from the Gradient Boosting PR curve,

trading some recall for much better accuracy:

TABLE IV: GB Threshold Tuning (holdout set)

Threshold	Accuracy	Recall	Precision	F1
0.5 (default)	0.509	0.772	0.135	0.229
0.646 (F1-opt.)	0.714	0.557	0.177	0.269

At the tuned threshold, accuracy reaches 0.714 and we still catch more than half of all true declines (recall 0.557). This is the recommended operating point for a monitoring tool.

D. Cross-Validation Stability

Expanding-window CV confirms the holdout results are consistent across time periods, not a lucky split:

TABLE V: Expanding-Window CV Results (4 folds, mean $\pm$ std)

Model	PR-AUC	ROC-AUC	Recall
Decision Tree	0.261 $\pm$ 0.086	0.737 $\pm$ 0.033	0.800 $\pm$ 0.093
Random Forest	0.321 $\pm$ 0.081	0.753 $\pm$ 0.026	0.760 $\pm$ 0.055
Gradient Boosting	0.302 $\pm$ 0.105	0.744 $\pm$ 0.041	0.799 $\pm$ 0.042

Random Forest has the highest mean CV PR-AUC (0.321) and the lowest standard deviation (0.081), making it the most stable model across different time periods. The decline rate per fold drops from 19.3% (2015 fold) to 8.8% (2018 fold), a sign of Spotify's retrospective downweighting bias discussed in Section V.

E. Feature Importance

Trajectory features—especially gap_from_peak, peak_popularity, and career_slope—ranked highest in both Random Forest and Gradient Boosting importance scores. Audio features (danceability, energy, valence) ranked lower. This suggests that *where an artist is in their career* matters more than *what their music sounds like* for predicting year-over-year decline in this dataset.

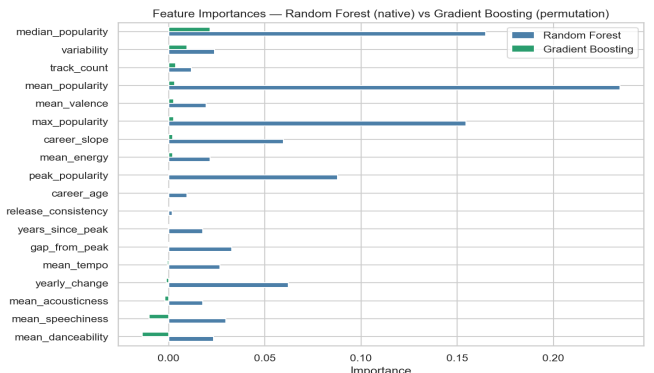


Fig. 2. Feature importances: RF (left) and GB permutation importance (right).

F. Three-Class Extension

We softened the binary label to three classes—declining ($\Delta \leq -10$), stable, growing ($\Delta \geq +10$). The ± 10 boundary absorbs noise into stable, giving the declining class 46% more training examples (608 $\rightarrow$ 889). Results:

TABLE VI: Three-Class Extension — Declining-Class PR-AUC

Model	PR-AUC [Decl.]	Baseline	Lift	Accuracy
Random Forest	0.320	0.165	1.94 $\times$	0.416
Gradient Boosting	0.288	0.165	1.75 $\times$	0.446

Declining-class PR-AUC improved from 0.201 to 0.320 for RF. Overall accuracy dropped (0.416) because three-way classification is harder—random chance is 33%, not 50%.

G. Lifecycle Clustering

We used K-Means ($k = 4$, selected by silhouette score) on artist-level career summaries after PCA (5 components, 90.2% variance). Four archetypes emerged:

TABLE VII: Lifecycle Archetypes (K-Means, $k = 4$)

Cluster	Name	n	Peak	Slope	Decline %
0	Niche Artists	276	36	+2.1/yr	4.8%
1	Mainstream Veterans	478	64	+0.2/yr	14.4%
2	Legacy Decliners	200	57	-17.7/yr	41.7%
3	Rising Stars	140	61	+31.2/yr	15.8%

Niche Artists (C0): Low popularity, stable, almost never decline. Intermittent releases. A small but consistent streaming footprint. **Mainstream Veterans (C1):** High historical peak, near-zero slope—plateaued but not collapsing. The largest and most stable group. **Legacy Decliners (C2):** Steep long-term drop, high variability, nearly half of all career years are declining. Partly amplified by Spotify's retrospective bias against older catalogs. **Rising Stars (C3):** Short careers, very steep upward slope, high final popularity—artists in early ascent.

Clusters are statistically associated with decline rates ($\chi^2 = 347.05$, $p < 0.0001$). C2 contains 100% high-decline artists; C0 only 15%. The silhouette score of 0.239 is modest—careers exist on a continuum, not in hard-boundary groups. Agglomerative clustering (Ward linkage) gave ARI = 0.585 vs. K-Means, confirming the groupings are not geometric artifacts.

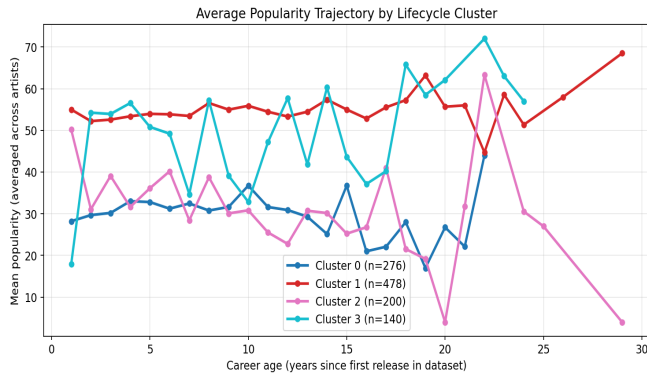


Fig. 3. Average popularity vs. career age per cluster. Legacy Decliners (C2) fall steeply; Rising Stars (C3) ascend rapidly.

V. HONEST ASSESSMENT AND LIMITATIONS

We want to be direct: **the results are below what we were hoping for**. A PR-AUC of 0.201 is 2.1× the random baseline, which proves the model is learning something real—but in a real-world setting, a label would need roughly 3–5× improvement before trusting a model with artist decisions. At our best threshold, precision is 0.177, meaning for every 10 artists the model flags as at risk, about 8 are false alarms. That is too noisy for automated decisions.

What held us back:

- **Small dataset.** We used only 32,000 tracks from one source. A companion dataset with 114,000 tracks across 114 genres was available but not incorporated. More data would likely improve performance noticeably.
- **No genre separation.** We trained one model for all artists. An EDM artist and a classical musician have very different career patterns. Separate models per genre would almost certainly perform better.
- **Spotify's retrospective bias.** Spotify weights recent streams more heavily than older ones. Artists from the 1990s appear to decline more steeply than they actually are, because their catalog loses algorithmic weight over time. This inflates our decline rate for older artists and confounds the model.
- **Annual aggregation loses detail.** We collapsed each year into one row per artist. A viral moment in December and a slow year look the same. This makes it hard to detect momentum shifts that happen mid-year.
- **Circular features.** Features like `gap_from_peak` and `career_slope` partly reflect that an artist is already declining—not purely predicting a future break. The model is better described as a monitoring tool than a true early-warning system.
- **Clustering is soft.** A silhouette score of 0.239 means the four archetypes overlap significantly. They are useful as narrative summaries of common career shapes, but not hard-boundary categories.

VI. WHAT WE WOULD DO DIFFERENTLY

If we had more time, these are the changes we believe would make the biggest difference, in priority order:

- **Merge the second dataset (highest priority).** The 114,000-track companion dataset would give 3.5× more data and 114 genres. More data is the single most likely improvement.
- **Build genre-specific models.** Train separate models for pop, rap, rock, EDM, etc. Genre fundamentally changes how careers evolve.
- **Use finer time resolution.** Aggregate at the quarter or month level instead of the year. This would expose within-year momentum shifts that annual aggregation currently hides.
- **Add external signals.** Social media follower growth, concert ticket sales, radio play, and press coverage all reflect career trajectory but are not in the Spotify dataset.
- **Try time-series clustering.** Instead of clustering on summary statistics, use Dynamic Time Warping (DTW) to cluster on actual trajectory shapes. Two artists with the same average slope but different patterns (steady decline vs. boom-bust) would then be correctly separated.
- **Prospective validation.** Our evaluation uses historical data. A real test would train on pre-2020 data, make predictions for 2020, and check outcomes in 2021–2022 without looking ahead.

VII. CONCLUSION

We built a full pipeline: artist-year aggregation, trajectory feature engineering, time-aware supervised decline prediction, and unsupervised lifecycle archetype discovery. The pipeline works and produces results that are statistically above chance. Our Random Forest achieves PR-AUC = 0.201 (2.1× baseline), and clustering reveals four interpretable archetypes validated by chi-square testing ($p < 0.0001$).

However, we are honest that these results are not strong enough for real deployment. The model would work as a rough

monitoring tool—flag the top artists for a human to review—but not as an automated decision system. The path to better results is clear: more data, genre stratification, and richer external signals.

REFERENCES

- [1] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [2] J. Davis and M. Goadrich, “The relationship between Precision-Recall and ROC curves,” in *Proc. 23rd ICML*, 2006, pp. 233–240.
- [3] O. Celma, *Music Recommendation and Discovery*. Springer, 2010.
- [4] S. Askin and M. Mauskopf, “What makes popular culture popular?” *Amer. Sociol. Rev.*, vol. 82, no. 5, pp. 910–944, 2017.
- [5] J. Singleton, “Spotify songs dataset,” Kaggle, 2020. <https://www.kaggle.com/datasets/joebeachcapital/30000-spotify-songs>
- [6] L. Breiman, “Random forests,” *Mach. Learn.*, vol. 45, pp. 5–32, 2001.
- [7] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. Springer, 2009.
- [8] S. M. Zackery, “The severity of the sophomore slump in the music industry,” Undergraduate thesis, Scripps College, 2014.